Cyber Security Overview

unit 1

1. Define cyber security. CO1 K1
2. List the elements of cyber security. CO1 K1
3. Define Cyber Crime CO1 K1
4. Illustrate CIA triad with a simple diagram. CO1 K2
5. List out the positive and negative impacts of internet in society. CO1 K1
6. Interpret the reasons for cyber crime. CO1 K2
7. Define DNS CO1 K1
8. Summarize espionage. CO1 K2
9. List the types of cyber criminals. CO1 K1
10. Compare Black hat, grey hat and white hat hackers CO1 K2
11. Classify cyber crimes. CO1 K2
12. What is cyber crime against individual?Give example. CO1 K2
13. What is cyber crime against property in short. Give example. CO1 K2
14. What is cyber crime against organization. Give example. CO1 K2
15. What is cyber crime against society. Give example. CO1 K2
16. Define cyber law with an example. CO1 K2
17. Classify cyber law with its types. CO1 K2
18. Outline the offenses and punishments in Indian IT act. CO1 K2
19. Summarize the importance of cyber security. CO1 K2
20. What is ARPANET in the development of the internet?

Here's a **neatly organized set of answers** for your Unit 1 questions:

1️⃣ Define cyber security.

Cyber security is the practice of protecting systems, networks, and programs from digital attacks, unauthorized access, or damage.

2️⃣ List the elements of cyber security.

- Application security
- Network security
- Information security
- Operational security
- End-user education
- Business continuity and disaster recovery

3️⃣ Define Cyber Crime.

Cyber crime refers to illegal activities carried out using computers or the internet, such as hacking, identity theft, or spreading malware.

4️⃣ Illustrate CIA triad with a simple diagram.

```lua
        Confidentiality
             /   \
            /     \
   Integrity ----- Availability
```

The **CIA triad** represents three key principles:

- **Confidentiality:** Keeping data private.
- **Integrity:** Ensuring data accuracy.
- **Availability:** Ensuring reliable access.

5️⃣ List out the positive and negative impacts of internet in society.

✅ Positive:

- Easy communication
- Access to information
- E-commerce and business growth

❌ Negative:

- Cyber bullying
- Privacy loss
- Spread of fake news

6️⃣ Interpret the reasons for cyber crime.

- Financial gain
- Political or ideological motives
- Revenge or personal grudges
- Desire for recognition or fame
- Weak security systems

7️⃣ Define DNS.

DNS (Domain Name System) is the system that translates human-readable domain names (like google.com) into IP addresses that computers use.

8️⃣ Summarize espionage.

Espionage is the act of spying or obtaining secret information without permission, often for political or commercial gain.

9️⃣ List the types of cyber criminals.

- Hackers
- Script kiddies
- Cyber terrorists
- Disgruntled employees
- Hacktivists
- Organized cybercriminal groups

🔟 Compare Black hat, grey hat, and white hat hackers.

Type	Activity
Black hat	Malicious hacking for personal gain
Grey hat	Unethical hacking but without harmful intent
White hat	Ethical hacking to improve security

1️⃣1️⃣ Classify cyber crimes.

- Against individuals
- Against property
- Against organizations
- Against society

1️⃣2️⃣ What is cyber crime against individual? Give example.

Crimes targeting individuals, like **phishing** or **identity theft**.
Example: Stealing someone's personal bank details.

1️⃣3️⃣ What is cyber crime against property in short? Give example.

Crimes targeting data or systems.

Example: Hacking into servers to steal intellectual property.

1️⃣4️⃣ What is cyber crime against organization? Give example.

Attacks on businesses or institutions.

Example: Distributed Denial of Service (DDoS) attack on a company's website.

1️⃣5️⃣ What is cyber crime against society? Give example.

Crimes causing social harm.

Example: Spreading fake news or child pornography.

1️⃣6️⃣ Define cyber law with an example.

Cyber law refers to legal regulations governing cyberspace activities.

Example: Indian IT Act, 2000.

1️⃣7️⃣ Classify cyber law with its types.

- Intellectual property law
- Data protection and privacy law
- Cybercrimes and offenses
- Electronic and digital signature laws

1️⃣8️⃣ Outline the offenses and punishments in Indian IT Act.

- **Offenses:** Hacking, data theft, identity theft, cyber terrorism.
- **Punishments:** Fines, imprisonment (up to life for cyber terrorism).

1️⃣9️⃣ Summarize the importance of cyber security.

Cyber security protects data, ensures privacy, builds trust, prevents financial loss, and safeguards critical infrastructure.

2️⃣0️⃣ What is ARPANET in the development of the internet?

ARPANET (Advanced Research Projects Agency Network) was the first packet-switching network and the foundation of today's internet, developed in the late 1960s.

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unit 2

1. Define OWASP. CO2 K1
2. List out OWASP Top 10 CO2 K1
3. How does a Trojan horse differ from other types of malicious
software.
CO2 K1
4. Summarize the scope of cyber attacks CO2 K2
5. Compare malicious attacks and threats. CO2 K2
6. Outline active threats with its types. CO2 K2
7. Describe security breach with a real time example CO2 K2
8. List out the types of malicious attacks CO2 K1
9. List the types of malwares with an example each. CO2 K2
10. What is malicious software. Give example. CO2 K2
11. Compare virus and worms CO2 K2
12. Outline common attack vectors in short. CO2 K2

13. What is social engineering attack? Give example. CO2 K2
14. What is wireless network attack? Give example. CO2 K2
15. What is web application attack? Give example. CO2 K2
16. Enumerate the types of wireless network attacks. CO2 K2
17. Outline the methods to prevent wireless network attacks. CO2 K1
18. List out the types of web application attacks CO2 K1
19.
What are the web application security strategies that organizations can implement to protect their web applications
CO2
K2
20. List the attack tools used by cyber criminals.

Here's your **Unit 2 answers summary** — clear and organized for easy revision:

1️⃣ Define OWASP.

OWASP (Open Web Application Security Project) is a non-profit organization focused on improving software security through community-led open-source projects.

2️⃣ List out OWASP Top 10.

- Injection
- Broken Authentication
- Sensitive Data Exposure
- XML External Entities (XXE)
- Broken Access Control
- Security Misconfiguration
- Cross-Site Scripting (XSS)
- Insecure Deserialization

- Using Components with Known Vulnerabilities
- Insufficient Logging and Monitoring

3️⃣ How does a Trojan horse differ from other types of malicious software?

A **Trojan horse** disguises itself as legitimate software but secretly performs malicious actions, unlike viruses or worms, which self-replicate.

4️⃣ Summarize the scope of cyber attacks.

Cyber attacks can target individuals, organizations, or governments, aiming to steal data, disrupt services, cause financial loss, or damage reputation.

5️⃣ Compare malicious attacks and threats.

Malicious Attack	Threat
Actual harmful activity	Potential risk or possibility
Example: launching malware	Example: existence of a vulnerability

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6️⃣ Outline active threats with its types.

Active threats involve direct actions to harm a system, such as:

- Denial of Service (DoS) attacks
- Man-in-the-Middle attacks
- Session hijacking
- Malware infection

7️⃣ Describe security breach with a real-time example.

A **security breach** is unauthorized access to data.

Example: 2017 Equifax breach where sensitive data of millions was exposed.

8️⃣ List out the types of malicious attacks.

- Malware attacks
- Phishing
- Denial of Service (DoS)
- Man-in-the-Middle
- SQL Injection
- Cross-Site Scripting (XSS)

9️⃣ List the types of malwares with an example each.

- Virus → Example: ILOVEYOU virus
- Worm → Example: Conficker worm
- Trojan → Example: Zeus Trojan
- Ransomware → Example: WannaCry
- Spyware → Example: CoolWebSearch

🔟 What is malicious software? Give example.

Malicious software (malware) is any software intentionally designed to cause harm.

Example: Ransomware encrypting files for ransom.

1️⃣1️⃣ Compare virus and worms.

Virus	Worm
Needs a host to spread	Spreads independently
Often attached to files	Self-replicates over networks

1️⃣2️⃣ Outline common attack vectors in short.

- Phishing emails
- Malicious websites
- Infected USB devices
- Unpatched software
- Weak passwords

1️⃣3️⃣ What is social engineering attack? Give example.

Social engineering manipulates people to reveal confidential info.
Example: Phishing emails tricking users to share passwords.

1️⃣4️⃣ What is wireless network attack? Give example.

Attacks targeting Wi-Fi networks.
Example: Eavesdropping on unsecured public Wi-Fi.

1️⃣5️⃣ What is web application attack? Give example.

Exploiting weaknesses in web apps.
Example: SQL Injection to steal database data.

1️⃣6️⃣ Enumerate the types of wireless network attacks.

- Eavesdropping
- Evil twin attack
- Rogue access point
- Denial of Service (DoS)
- Replay attacks

1️⃣7️⃣ Outline the methods to prevent wireless network attacks.

- Use WPA3 encryption
- Set strong Wi-Fi passwords
- Disable SSID broadcasting
- Use MAC address filtering
- Regularly update firmware

1️⃣8️⃣ List out the types of web application attacks.

- SQL Injection
- Cross-Site Scripting (XSS)
- Cross-Site Request Forgery (CSRF)
- File Inclusion attacks
- Broken Authentication

1️⃣9️⃣ What are the web application security strategies that organizations can implement to protect their web applications?

- Input validation and sanitization

- Regular vulnerability testing
- Using secure authentication mechanisms
- Keeping software updated
- Implementing security headers

2️⃣0️⃣ List the attack tools used by cyber criminals.

- Metasploit
- Nmap
- Wireshark
- John the Ripper
- Aircrack-ng
- SQLmap
- Hydra

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unit 3

1. Summarize the purpose of harvester in short. CO3 K2
2. Define Whois with an example. CO3 K1
3. List the key aspects of Netcrafts. CO3 K1
4. Mention the key tools for host reconnaissance CO3 K1
5. Summarize the key techniques in social engineering reconnaissance. CO3 K2
6. Classify the technique that involve leaving infected USB drives in public places or offering free downloads that contain malware. CO3 K2

7.
Illustrate the steps taken by organisations to protect against
social engineering attacks by reducing their susceptibility during the reconnaissance phase
CO3 K2
8. Mention the types of scanning in reconnaissance. CO4 K1
9. Compare network scanning and vulnerability scanning. CO4 K2
10. Relate ping scan, SyNC scan and XMAS scan. CO4 K2
11.
Interpret the categories of port scan that reveal the status of
network or server.
CO4 K2
12. Mention the purpose of network scanning. CO4 K1
13. Compare credentialed scan and non-credentialed scan. CO4 K2
14. Enumerate the steps for scanning methodologies. CO4 K2
15.
How can banner grabbing be useful for identifying the services running on a target system?
CO4 K1
16. List important flags in Nmap switches. CO4 K1
17. Relate NULL scan to XMAS and FIN CO4 K2
18. Mention the limitations of XMAS and NULL scan CO4 K1
19.
How can organizations protect against banner grabbing to avoid revealing sensitive information?
CO4 K1
20.
List the important elements to determine the operating system in fingerprinting.

Here's your **Unit 3 answers summary**, neatly organized for revision and quick study:

1️⃣ Summarize the purpose of harvester in short.

The **Harvester** is a tool used to gather email addresses, subdomains, and hostnames from public sources for reconnaissance.

2️⃣ Define Whois with an example.

Whois is a protocol that provides domain registration details, like the owner, registrar, and contact info.
Example: `whois example.com`

3️⃣ List the key aspects of Netcraft.

- Website profiling
- Hosting details
- Server technology identification
- SSL certificate information
- Historical uptime reports

4️⃣ Mention the key tools for host reconnaissance.

- Nmap
- Netcat
- The Harvester
- Maltego
- Whois

5️⃣ Summarize the key techniques in social engineering reconnaissance.

- Pretexting (creating a false scenario)

- Phishing (sending deceptive messages)
- Dumpster diving (retrieving discarded sensitive info)
- Shoulder surfing (watching over someone's shoulder)
- Baiting (offering something to entice victims)

6️⃣ Classify the technique that involves leaving infected USB drives in public places or offering free downloads that contain malware.

This is called **baiting** — a type of social engineering attack.

7️⃣ Illustrate the steps taken by organisations to protect against social engineering attacks by reducing their susceptibility during the reconnaissance phase.

- Employee training on social engineering risks
- Implementing strict security policies
- Limiting publicly available sensitive information
- Using strong authentication methods
- Regularly testing employees with simulated attacks

8️⃣ Mention the types of scanning in reconnaissance.

- Port scanning
- Network scanning
- Vulnerability scanning

9️⃣ Compare network scanning and vulnerability scanning.

Network Scanning	Vulnerability Scanning
Identifies active devices and open ports	Identifies security weaknesses in systems
Focused on network structure	Focused on system vulnerabilities and patches

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🔟 Relate ping scan, SYN scan, and XMAS scan.

- **Ping scan:** Checks if hosts are alive.
- **SYN scan:** Half-open scan to detect open ports.
- **XMAS scan:** Sends TCP packets with FIN, URG, and PUSH flags to detect port states.

1️⃣1️⃣ Interpret the categories of port scan that reveal the status of network or server.

- Open scan
- Closed scan
- Filtered scan
- Unfiltered scan

1️⃣2️⃣ Mention the purpose of network scanning.

To discover live hosts, open ports, and services running on a network to understand its topology and identify potential attack points.

1️⃣3️⃣ Compare credentialed scan and non-credentialed scan.

Credentialed Scan	Non-Credentialed Scan
Uses valid credentials for deeper access	No credentials; only external view

Credentialed Scan	Non-Credentialed Scan
Provides detailed vulnerability insights	Limited visibility; surface-level details

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1️⃣4️⃣ Enumerate the steps for scanning methodologies.

1. Identify active hosts
2. Scan open ports
3. Detect services and versions
4. Identify vulnerabilities
5. Document findings

1️⃣5️⃣ How can banner grabbing be useful for identifying the services running on a target system?

Banner grabbing collects service banners, revealing software versions and types, which helps identify potential vulnerabilities.

1️⃣6️⃣ List important flags in Nmap switches.

- `-sS` → SYN scan
- `-sT` → TCP connect scan
- `-sU` → UDP scan
- `-O` → OS detection
- `-A` → Aggressive scan (OS + version + scripts)

1️⃣7️⃣ Relate NULL scan to XMAS and FIN.

All three send unusual TCP packets:

- **NULL scan:** No flags set
- **FIN scan:** Only FIN flag set
- **XMAS scan:** FIN, URG, and PUSH flags set
  They all aim to bypass some firewalls and reveal port states.

1️⃣8️⃣ Mention the limitations of XMAS and NULL scan.

- Work only on specific systems (like UNIX)
- Ineffective against modern firewalls
- Do not work on Windows (which replies the same for all)

1️⃣9️⃣ How can organizations protect against banner grabbing to avoid revealing sensitive information?

- Disable unnecessary banners
- Use generic or custom banners
- Apply firewall filtering
- Keep systems updated

2️⃣0️⃣ List the important elements to determine the operating system in fingerprinting.

- TTL (Time To Live) values
- TCP window size
- TCP/IP stack behavior
- ICMP responses
- TCP options (like MSS, SACK)

✅ If you want, I can **compile Units 1, 2, and 3 into a complete printable notes file or PDF** — just let me know! 🌟📘

unit 4

1. What is Intrusion Detection System? CO5 K1
2. Classify Intrusion Detection System. CO5 K2
3. Differentiate Network and Host intrusion detection system. CO5 K2
4. Mention the Detection Method of Intrusion Detection System. CO5 K1
5. List the categories of Host-Based Intrusion Detection Systems CO5 K1
6. What are the limitations and challenges of Host-Based Intrusion Detection Systems. CO5 K1
7. Classify the different detection methods to identify suspicious traffic and abnormal behaviour in network Intrusion Detection Systems CO5 K2
8. Compare the advantages and limitations of network Intrusion Detection Systems. CO5 K2
9. Abbreviate IDMEF. CO5 K2
10. Mention the key characteristics of Distributed Intrusion Detection Systems. CO5 K1
11. Relate the advantages and disadvantages of Distributed Intrusion Detection Systems. CO5 K2
12. Define honeypot with an example. CO5 K1
13. Classify honeypot based on deployment and interaction. CO5 K2
14. Differentiate advantages and disadvantages of honeypot. CO5 K2
15. Summarize SNORT briefly. CO5 K2

16. What Are the Different SNORT Modes? CO5 K1
17. Illustrate the Uses of SNORT Rules. CO5 K2
18.
Differentiate Normal Internet Activities and Malicious Activities by SNORT
CO5 K2
19. Mention the features of SNORT. CO5 K1
20. Classify honeypot by the type of activity they detect.

Here's your **Unit 4 answers summary**, clean and exam-ready:

1️⃣ What is Intrusion Detection System?

An **Intrusion Detection System (IDS)** monitors network or system activities for malicious actions or policy violations.

2️⃣ Classify Intrusion Detection System.

- Network-based IDS (NIDS)
- Host-based IDS (HIDS)
- Hybrid IDS

3️⃣ Differentiate Network and Host Intrusion Detection System.

Network IDS (NIDS)	Host IDS (HIDS)
Monitors network traffic	Monitors activity on a specific host
Deployed at network boundaries	Installed on individual machines
Detects external attacks	Detects internal misuse or file changes

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4️⃣ Mention the Detection Method of Intrusion Detection System.

- Signature-based detection
- Anomaly-based detection
- Hybrid detection

5️⃣ List the categories of Host-Based Intrusion Detection Systems.

- File integrity monitoring
- Log analysis
- System call monitoring

6️⃣ What are the limitations and challenges of Host-Based IDS?

- High resource usage
- Limited to individual hosts
- Hard to manage in large networks
- Can be disabled if the host is compromised

7️⃣ Classify the different detection methods in Network IDS.

- Signature-based detection
- Anomaly-based detection
- Stateful protocol analysis

8️⃣ Compare the advantages and limitations of Network IDS.

Advantages	Limitations
Monitors entire network traffic	May struggle with encrypted traffic
Detects external attacks early	High false positives
Centralized monitoring	Needs dedicated hardware for high speed

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9️⃣ Abbreviate IDMEF.

IDMEF → Intrusion Detection Message Exchange Format

🔟 Mention the key characteristics of Distributed IDS.

- Multiple IDS sensors across locations
- Centralized analysis and correlation
- Scalability for large networks
- Cooperative detection

1️⃣1️⃣ Relate the advantages and disadvantages of Distributed IDS.

Advantages	Disadvantages
Broader coverage across network	Complex deployment
Coordinated attack detection	High resource consumption
Scalable and flexible	Management and maintenance challenges

1️⃣2️⃣ Define honeypot with an example.

A **honeypot** is a decoy system designed to attract attackers and study their behavior.
Example: A fake server set up to log unauthorized access attempts.

1️⃣3️⃣ Classify honeypot based on deployment and interaction.

- **Deployment:** Production honeypots, Research honeypots
- **Interaction level:** Low-interaction, Medium-interaction, High-interaction

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1️⃣4️⃣ Differentiate advantages and disadvantages of honeypot.

Advantages	Disadvantages
Captures real attack data	Can be discovered by skilled attackers
Diverts attackers away from real assets	Needs constant monitoring
Helps improve defenses	Risk of being used to attack others

1️⃣5️⃣ Summarize SNORT briefly.

SNORT is an open-source network intrusion detection and prevention system (NIDS/NIPS) that analyzes traffic in real time using rules and signatures.

1️⃣6️⃣ What Are the Different SNORT Modes?

- Sniffer mode (packet capture)
- Packet logger mode (logs traffic)
- NIDS mode (analyzes and detects attacks)

1️⃣7️⃣ Illustrate the Uses of SNORT Rules.

SNORT rules define traffic patterns or conditions to detect attacks or suspicious activities (e.g., blocking SQL injections or port scans).

1️⃣8️⃣ Differentiate Normal Internet Activities and Malicious Activities by SNORT.

SNORT uses rules to flag abnormal patterns, like:

- **Normal:** Standard HTTP requests, DNS queries
- **Malicious:** Port scans, exploit attempts, unusual payloads

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1️⃣9️⃣ Mention the features of SNORT.

- Real-time packet analysis
- Protocol analysis
- Signature-based detection
- Logging and alerting
- Extensible with plugins

2️⃣0️⃣ Classify honeypot by the type of activity they detect.

- Malware honeypots (detect malware infections)
- Spam honeypots (detect spam sources)
- Database honeypots (detect database attacks)
- Web honeypots (detect web application attacks)

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unit 5

1. Illustrate the primary need for firewalls in network security. CO6 K2
2. Demonstrate the key characteristics of a firewall. CO6 K2
3. Describe access policy in the context of firewalls. CO6 K2
4. List the types of firewalls. CO6 K1
5. Interpret packet-filtering firewall and its functions. CO6 K2
6. Differentiate stateful inspection firewall from a stateless firewall. CO6 K2
7. Summarize application-level gateway firewall in short. CO6 K2
8. Define firewall basing with an example. CO6 K1
9. Compare host-based and network-based firewalls. CO6 K2
10. Interpret the location of firewall in a network for optimal security
with a neat diagram.
CO6 K1
11. Relate DMZ (Demilitarized Zone) to firewall configurations. CO6 K2
12. Illustrate a single-homed firewall setup with an example. CO6 K2
13. Infer how a dual-homed firewall enhance security. CO6 K2
14. What is the primary purpose of an Intrusion Prevention System
(IPS)?
CO6 K1
15. Relate Intrusion Prevention System (IPS) from Intrusion Detection
System (IDS)
CO6 K2
16. List the types of IPS based on deployment. CO6 K1
17. What is a host-based IPS (HIPS), and where is it deployed? CO6 K1
18. What is the role of a network-based IPS (NIPS)? CO6 K1
19. Outline the significance of Unified Threat Management

(UTM). CO6 K2
20. Mention two example products that offer Unified Threat Management(UTM) solutions

Here's your **Unit 5 answers summary** — clean, pointwise, and exam-ready:

1️⃣ Illustrate the primary need for firewalls in network security.

Firewalls act as barriers that control incoming and outgoing traffic, protecting networks from unauthorized access, malware, and cyberattacks.

2️⃣ Demonstrate the key characteristics of a firewall.

- Filters traffic based on rules
- Logs traffic and events
- Provides access control
- Supports VPN integration
- Stateful or stateless inspection

3️⃣ Describe access policy in the context of firewalls.

An access policy defines which types of traffic are allowed or denied through the firewall, based on rules like IP address, port, and protocol.

4️⃣ List the types of firewalls.

- Packet-filtering firewall
- Stateful inspection firewall
- Application-level gateway (proxy firewall)
- Circuit-level gateway

- Next-generation firewall (NGFW)

5️⃣ Interpret packet-filtering firewall and its functions.

A packet-filtering firewall checks individual packets' headers (source, destination, port, protocol) and allows or blocks them based on predefined rules.

6️⃣ Differentiate stateful inspection firewall from a stateless firewall.

Stateful Firewall	Stateless Firewall
Tracks connection state	Inspects packets independently
More secure and context-aware	Faster but less intelligent
Can detect abnormal traffic patterns	Only enforces simple rules on each packet

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7️⃣ Summarize application-level gateway firewall in short.

An application-level gateway (proxy firewall) filters traffic at the application layer, inspecting content and enforcing protocol-specific security.

8️⃣ Define firewall basing with an example.

Firewall basing refers to where the firewall is placed:
Example: Placing a firewall on a dedicated hardware device or as software on a server.

9️⃣ Compare host-based and network-based firewalls.

Host-Based Firewall	Network-Based Firewall
Runs on individual devices	Protects the entire network perimeter
Controls traffic to/from that host	Controls traffic between networks
Limited scope	Broad network-level protection

🔟 Interpret the location of firewall in a network for optimal security (with diagram).

The firewall is placed between the internal network and the external network (internet), often alongside a DMZ for public-facing services.
(Diagram usually shows: Internal Network → Firewall → Internet)

1️⃣1️⃣ Relate DMZ (Demilitarized Zone) to firewall configurations.

A **DMZ** is a network segment between two firewalls (or between internal and external networks) that hosts public-facing services like web or mail servers, isolating them from internal systems.

1️⃣2️⃣ Illustrate a single-homed firewall setup with an example.

A **single-homed firewall** has one network interface connected between the private network and the internet, acting as a basic filtering point.

1️⃣3️⃣ Infer how a dual-homed firewall enhances security.

A **dual-homed firewall** has two network interfaces (one for internal, one for external), providing better isolation and control between trusted and untrusted networks.

1️⃣4️⃣ What is the primary purpose of an Intrusion Prevention System (IPS)?

An **IPS** monitors network traffic and actively blocks or prevents identified threats in real-time.

1️⃣5️⃣ Relate Intrusion Prevention System (IPS) to Intrusion Detection System (IDS).

IPS	IDS
Prevents and blocks malicious traffic	Detects and alerts on suspicious traffic
Active defense	Passive monitoring

1️⃣6️⃣ List the types of IPS based on deployment.

- Network-based IPS (NIPS)
- Host-based IPS (HIPS)
- Wireless IPS (WIPS)
- Network behavior analysis (NBA)

1️⃣7️⃣ What is a host-based IPS (HIPS), and where is it deployed?

A **HIPS** is installed on individual devices (hosts) to monitor and block suspicious activities directly on the host system.

1️⃣8️⃣ What is the role of a network-based IPS (NIPS)?

A **NIPS** monitors and filters network traffic at the perimeter, preventing malicious activity before it reaches internal systems.

1️⃣9️⃣ Outline the significance of Unified Threat Management (UTM).

UTM integrates multiple security functions (firewall, IDS/IPS, antivirus, VPN, content filtering) into a single device, simplifying management and improving overall protection.

2️⃣0️⃣ Mention two example products that offer Unified Threat Management (UTM) solutions.

- Fortinet FortiGate
- Sophos XG Firewall

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